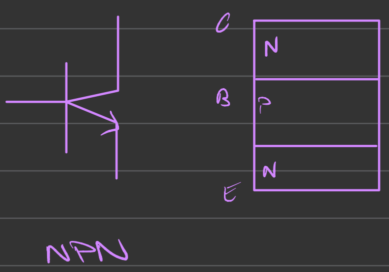
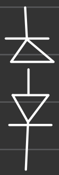
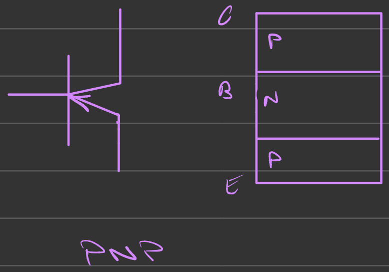
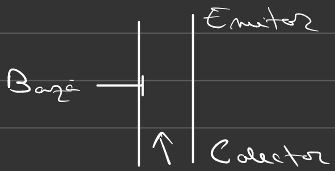
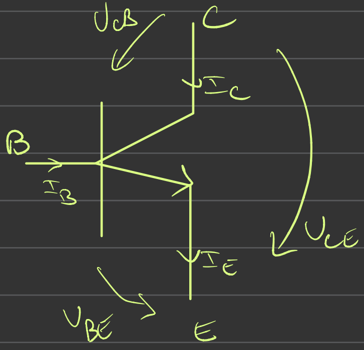
Curs 6



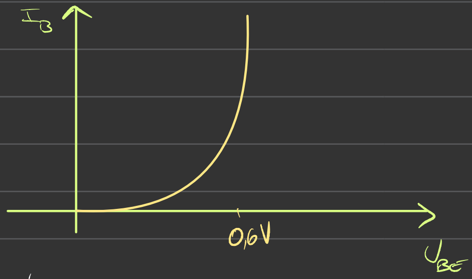


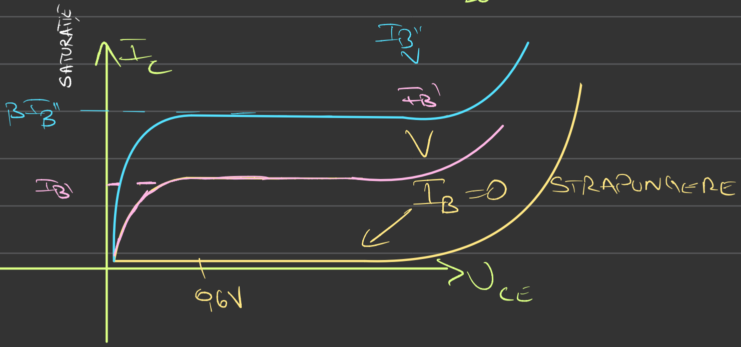






Caracteristică statică





$P_D \simeq U_{CE} I_C$ (puterea disipată)

RAN = regiunea activă normală (i inversă)

în RAN → $U_{BE} > 0$ $U_{CB} > 0$ (NPN) → direct
↘ $U_{EB} > 0$ $U_{BC} > 0$ (PNP) → invers

în SAT → $U_{BE} > 0$ $U_{CB} < 0$ (NPN)
↓ $U_{EB} > 0$ $U_{BC} < 0$ (PNP)

în RAI → $U_{BE} < 0$ $U_{CB} < 0$ (NPN)
↓ $U_{EB} < 0$ $U_{BC} < 0$ (PNP)

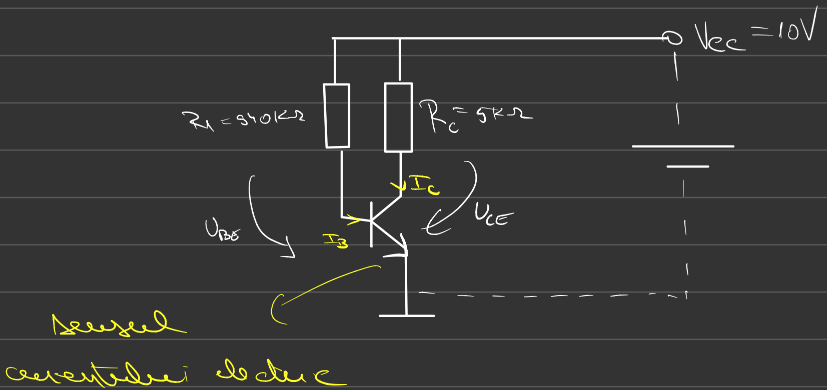
în BLOCAT → $U_{BE} < 0$ $U_{CB} > 0$ (NPN) ($I_C = 0$)
↓ $U_{EB} < 0$ $U_{BC} > 0$ (PNP)

$I_C = \beta I_B$ → β = zeci/sute (10–100)

$U_{BE} \simeq 0,6V$

Polarizarea tranzistorului

→ Cum să ținem tranzistorul în RAN



$(\beta = 100)$

Pp că T RAN ⇒ $I_C = \beta I_B$ $U_{BE} \simeq 0,6V$

Punct static de funcționare

I_C U_{CE}

⇒ $V_{CC} = R_1 I_B + U_{BE}$

$$I_B = \frac{V_{CC} - U_{BE}}{R_1} \simeq \frac{10 - 0,6}{940} = 0,01 \text{ mA}$$

$I_C = \beta I_B = 1 \text{ mA}$

$V_{CC} = I_C \cdot R_C + U_{CE}$ ⇒ $U_{CE} = 10 - 1\text{ mA} \cdot 5k\Omega = 5V$

⇒ $U_{CE} = 5V > U_{BE} = 0,6V$

Dacă $\beta = 200$

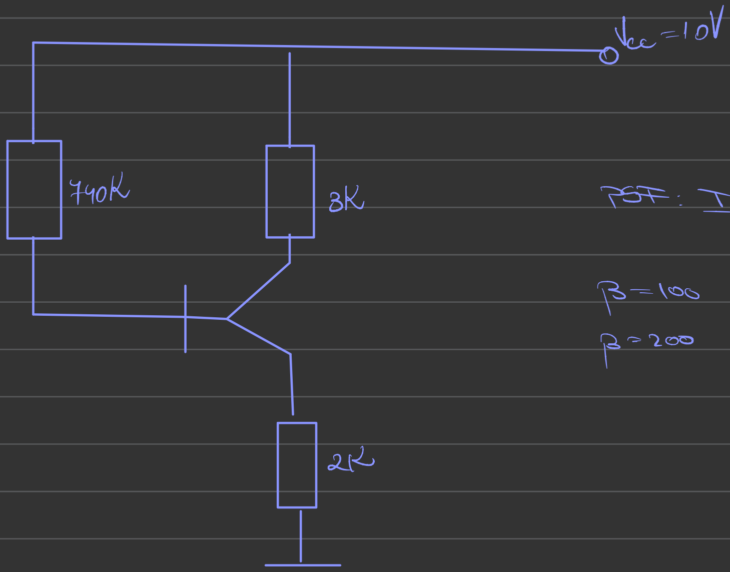
$I_B = 0{,}01\ mA$

$I_C = 2\ mA$

$U_{CE} = 0$ nu mai este în RAN ($< 0{,}6\ V$)

Dacă $\beta = 500$

$\Rightarrow U_{CE} = 0$



PSF: I_C U_{CE}

$\beta = 100$

$\beta = 200$